
Bachelor of Information Technology(BIT)

Course Title: BIT 353

Management Information System

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Course Contents

Our Recommended Text books:

1. Management Information Systems: Managing the Digital Firm, Kenneth C. Laudon and Jane P. Laudon, 17th Edition, Pearson, 2022.
2. Management Information Systems, James O'Brien and George Marakas, 10th Edition, McGraw Hill.

Course Contents

Unit 1: Information Systems in Global Business Today (3 Hrs.)

- What's New in Management Information System?
- Globalization Challenges and Opportunities: A Flattened World, The Emerging Digital Firm, Strategic Business Objectives of Information System, Information System and its Dimensions, Business Perspective on Information Systems, Complementary Assets, Academic Disciplines used to Study Information Systems, Sociotechnical Systems

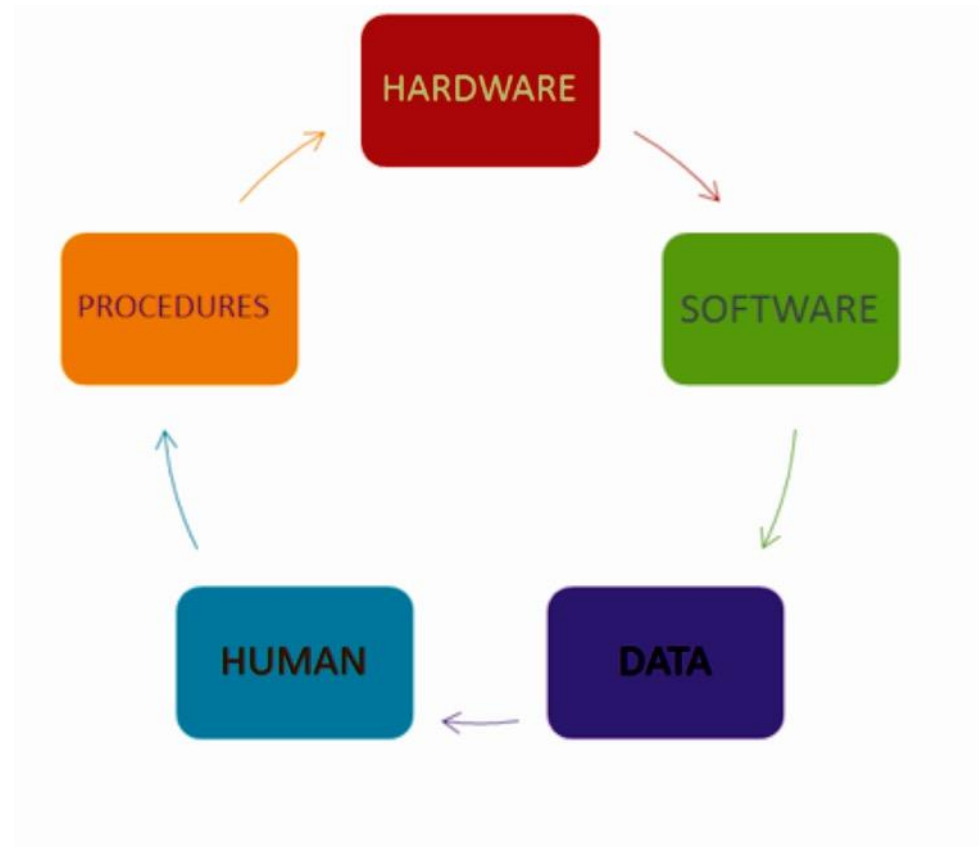
Information Systems in Global Business Today

An Information system that is the sub system of the business. Especially, it is an arrangement of interdependent of human and machine components that interact to support the operation when organization needed decision making.

The components of Information systems are

- **People, hardware, software, data and networks** are the five basic resources of information systems.

Resources of Information System



Types of Information System

1. Strategic Information System
2. Operational Information System
3. Tactical or MIS

Strategic Information System

This information is required by top most management for long range planning policies for next few years. As for example, financial investment, population growth and human resources etc. A strategic Information system(SIS) is a system that helps companies change their business, strategy and structure.

- Decision support system(DSS)
- Artificial Intelligence(AI)
- Expert System(ES)

Strategic Information System

- **Decision support system(DSS)**

A DSS is an information system that support business or organizational decision-making activities. A decision support system(DSS) is a computer based application that collects, organizes and analyzes business data to facilitate quality business decision-making for management, operations and planning.

- **Artificial Intelligence(AI)**

AI is the branch of computer science whose goal is to develop a machine with the human intelligence.

Strategic Information System

Four goals of AI

- System think like human.
- System act like human.
- System think rationally.
- System act rationally.

Strategic Information System

- Expert System(ES)

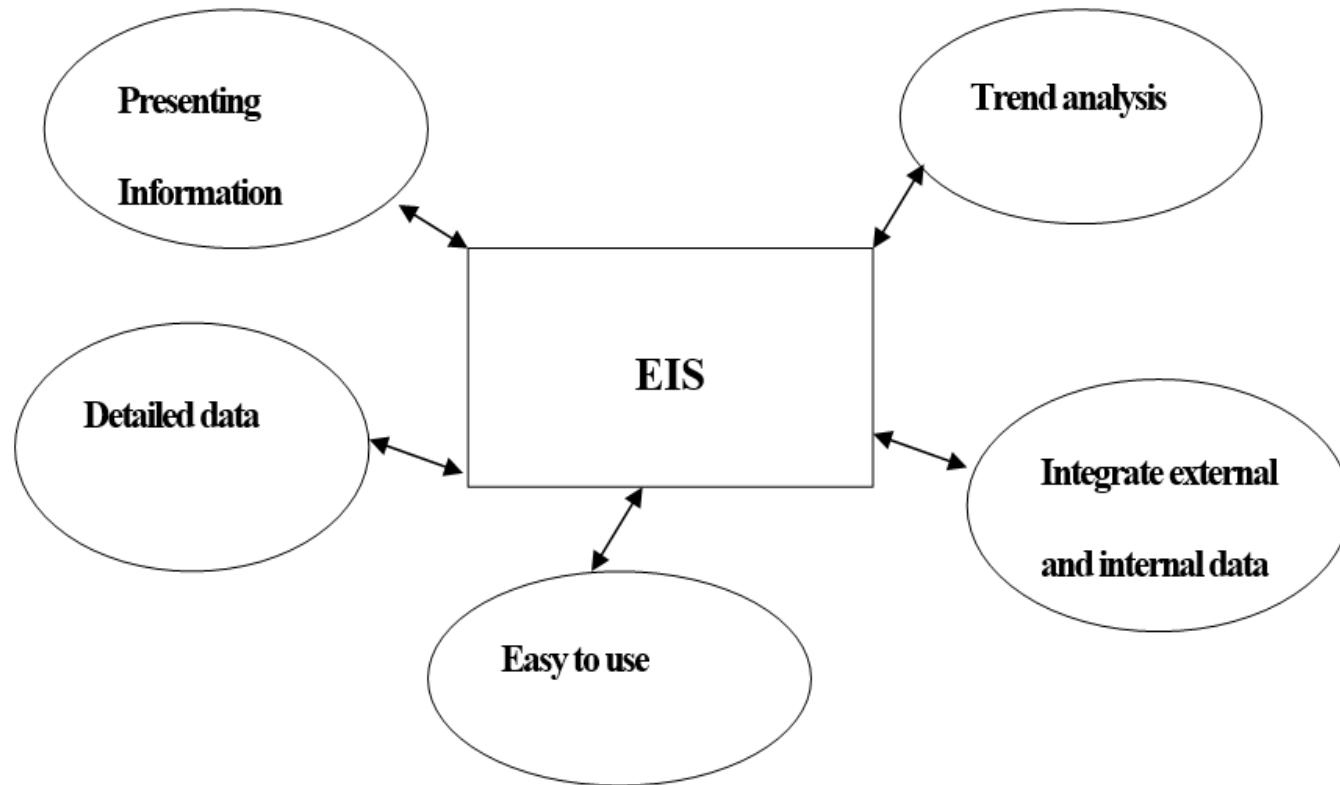
An expert system is a computer based system that uses AI technologies to simulate the judgment and behavior of a human or an organization that has expert knowledge and experience in a particular field.

Strategic Information System

- Executive Information System(EIS)

An **executive information system (EIS)** also known as an **executive support system (ESS)** is a type of management support system that facilitates and supports senior executive information and decision-making needs. It provides easy access to internal and external information relevant to organizational goals. It is commonly considered a specialized form of decision support system(DSS).

Strategic Information System



Operational Information System

This type of information is required by low management for daily and short term planning. This system can be used day by day activities of an organization. As for example, keeping employee attendance activities, overdue purchase and current stocks available etc.

- Transaction Processing System(TPS)
- Office Automation System(OAS)

Operational Information System

- Transaction Processing System(TPS)

A transaction processing system(TPS) is a set of information which processes the data transaction in database system that monitors transaction programs. A TPS is a software system that supports transaction processing.

Operational Information System

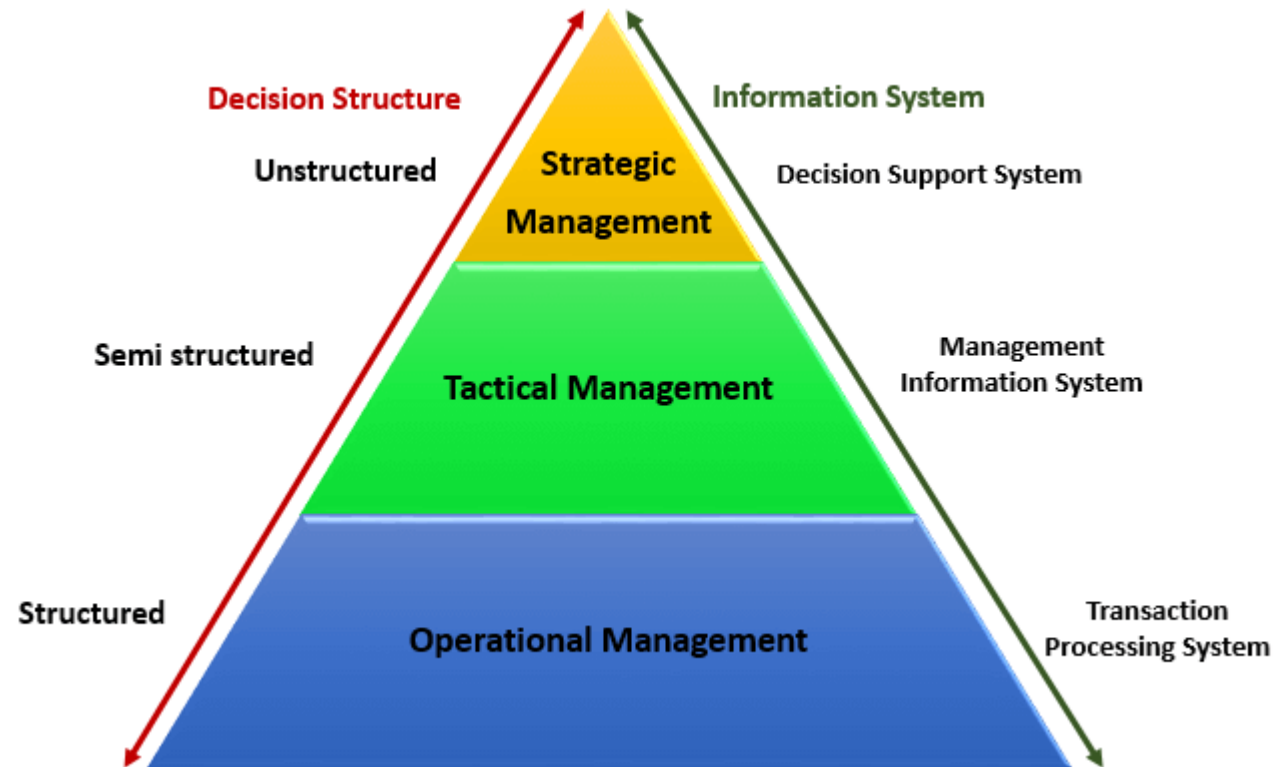
- Office Automation System(OAS)

OAS are computer based information systems that collect,process,store and transmit electronic messages, documents among individuals, work groups and organizations.

Tactical or MIS

This type of information system is needed to take short term decision to run the organization efficiently. They assist managers for collecting, storing, managing and retrieving information for organization needed. As for example, statistical summaries, comparative analysis, projections, periodic reports etc.

Pyramid Structure of IS



Management Information System(MIS)

Management consists of the interlocking functions of creating corporate policy and organizing, planning, controlling and directing an organization's resources in order to achieve the objectives. A **system** is a collection of elements or components that are organized for a common purpose.

Management Information System(MIS)

MIS provides:

- To the right information
- At the right place
- At the right time
- In the right form
- To the right person

Role of Information System in business Today

The environment in which organizations operate has changed dramatically over the last few years. Increased competition, globalization, the influence of the Internet and international events affect the performance and survival of organizations on a world-wide scale. The Internet has changed the way organizations do business, from the acquisition and servicing of customers to the management of their relations with suppliers. This is not only revolutionizing the way people access information, communicate, shop and entertain themselves but also the way organizations compete and operate.

Role of Information System in business Today

As the cloud continues to gain momentum, companies worldwide are increasing their spending towards the IT sector. Traditional IT systems are declining as modern digital technologies like AI and virtual reality are proving to be strong business driving forces. Indeed, the IT sector is being dominated by emerging cloud infrastructure, AI, VR, block chain and other modern technologies.

The Three Fundamental Roles of Information Systems in Business

- Operations

A business information system carries out specific functions in support of operations, including payroll, employee record storage, preparing and storing company documents and credit card processing. If your company operates a manufacturing line, the information system can schedule tasks and processes while keeping quality records.

The Three Fundamental Roles of Information Systems in Business

Some companies such as graphic designers or advertising companies use the information system to produce their products and services. In supporting operations, the information system can increase efficiency and improve employee productivity.

The Three Fundamental Roles of Information Systems in Business

- Controls

Monitoring and controlling the activities of employees is a core function of information systems. This is especially true of financial transactions in which management has a duty to prevent fraud and theft. In this role, the security of the information system is critical, as managers rely on it to track payments received from customers and invoices from suppliers.

The Three Fundamental Roles of Information Systems in Business

- Decisions

A third fundamental role for information systems is management support in making decisions. Managers can get all the information they need about company activities from the system. Complete, accurate information means management decisions are more effective. More sophisticated systems can go beyond supplying data to running scenarios:

Strategic Business Objectives of Information System

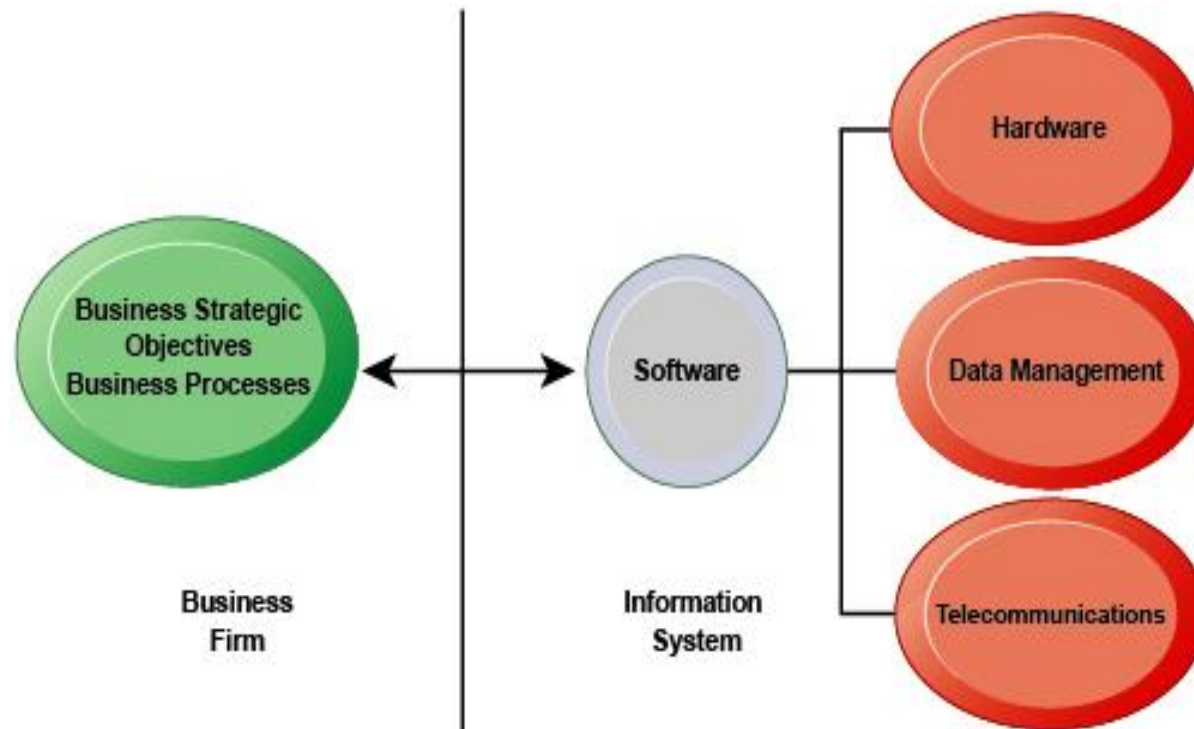
- Think local and act globally.

Although many managers are familiar with the reasons why managing their typical resources such as equipment and people are important, it is worthwhile to take a moment to examine the growing interdependence between a firm's ability to use information technology and its ability to implement corporate strategies and achieve corporate goals. Specifically, business firms invest heavily in information to achieve six strategic business objectives:

Strategic Business Objectives of Information System

- Operational excellence
- New products, services and business models
- Customer and supplier intimacy
- Improved decision making
- Competitive advantage
- Survival

Strategic Business Objectives of Information System



New in Management Information System(MIS)

A **management information system (MIS)** is an information system used for decision-making and for the coordination, control, analysis and visualization of information in an organization.

The study of the management information systems involves people, processes and technology in an organizational context.

New in Management Information System(MIS)

In a corporate setting, the ultimate goal of the use of a management information system is to increase the value and profits of the business. This is done by providing managers with timely and appropriate information allowing them to make effective decisions within a shorter period of time.

History of MIS

While it can be contested that the history of management information systems date as far back as companies using ledgers to keep track of accounting, the modern history of MIS can be divided into five eras originally identified by Kenneth C. Laudon and Jane Laudon in their seminal textbook Management Information Systems.

History of MIS

- First Era – Mainframe and minicomputer computing
- Second Era – Personal computers
- Third Era – Client/server networks
- Fourth Era – Enterprise computing
- Fifth Era – Cloud computing

What Are the Benefits of Globalization?

Globalization impacts businesses in many different ways. But those who decide to take on international expansion find several benefits, including:

1. Access to New Cultures

Globalization makes it easier than ever to access foreign culture, including food, movies, music, and art. This free flow of people, goods, art and information is the reason you can have Thai food delivered to your apartment as you listen to your favorite UK-based artist or stream a Bollywood movie.

What Are the Benefits of Globalization?

2. The Spread of Technology and Innovation

Many countries around the world remain constantly connected, so knowledge and technological advances travel quickly. Because knowledge also transfers so fast, this means that scientific advances made in Asia can be at work in the United States in a matter of days.

What Are the Benefits of Globalization?

3. Lower Costs for Products

Globalization allows companies to find lower-cost ways to produce their products. It also increases global competition which drives prices down and creates a larger variety of choices for consumers. Lowered costs help people in both developing and already-developed countries live better on less money.

What Are the Benefits of Globalization?

4. Higher Standards of Living Across the Globe

Developing nations experience an improved standard of living—thanks to globalization. According to the World Bank , extreme poverty decreased by 35% since 1990. Further, the target of the first Millennium Development Goal was to cut the 1990 poverty rate in half by 2015. This was achieved five years ahead of schedule, in 2010. Across the globe, nearly 1.1 billion people have moved out of extreme poverty since that time.

What Are the Benefits of Globalization?

5. Access to New Markets

Businesses gain a great deal from globalization, including new customers and diverse revenue streams. Companies interested in these benefits look for flexible and innovative ways to grow their business overseas. International Professional Employee Organizations(PEOs) make it easier than ever to employ workers in other countries quickly and compliantly. This means that for many companies, there is no longer the need to establish a foreign entity to expand overseas.

The Challenges of Globalization

While globalization offers many benefits, it's not without challenges. Velocity Global's 2020 state of Global Expansion™ Report: Technology Industry reveals some of the top challenges that U.S. and UK tech leaders face when taking their companies global and leaders of other companies likely face the same obstacles.

The Challenges of Globalization

1. International Recruiting

It's not surprising that 30% of U.S. and UK tech leaders cited international recruiting their most common challenge. Recruiting across borders creates unknowns for HR teams. First, companies create a plan for how they will interview and thoroughly vet candidates to make sure they are qualified when thousands of miles separate them from headquarters. Next, companies need to know the market's demands for salaries and benefits to make competitive offers. To ensure successful hires, HR teams must factor in challenges like time zones, cultural differences and language barriers to find a good fit for the company.

The Challenges of Globalization

2. Managing Employee Immigration

Immigration challenges cause a lot of headaches internally, which is why 28% of U.S. and UK tech leaders agreed it was one of their top challenges. Immigration laws change often, and in some countries, it is extremely difficult to secure visas for employees that are foreign nationals.

The Challenges of Globalization

3. Loss of Cultural Identity

While globalization has made foreign countries easier to access. It has also begun to meld unique societies together. The success of certain cultures throughout the world caused other countries to emulate them. But when cultures begin to lose their distinctive features, we lose our global diversity.

The Challenges of Globalization

4. Foreign Worker Exploitation

Lower costs do benefit many consumers, but it also creates tough competition that leads some companies to search for cheap labor sources. Some western companies ship their production overseas to countries like China and Malaysia.

The Challenges of Globalization

5. Global Expansion Difficulties

For businesses that want to go global and discover the benefits of globalization, setting up a compliant overseas presence is difficult. If companies take the traditional route of setting up an entity, they need substantial upfront capital, sometimes up to \$ 20,000 and costs of \$200,000 annually to maintain the business. Additionally, global businesses must keep up with different and ever-changing labor laws in new countries. When expanding into new countries, companies must be aware of how to navigate new legal systems.

The Challenges of Globalization

6. Immigration Challenges and Local Job Loss

The political climates in the United States and Europe show that there are different viewpoints on the results of globalization. Many countries around the globe are tightening their immigration rules, and it is harder for immigrants to find jobs in new countries.

How Globalization Changes Your Daily Businesses Operations

Both the benefits and challenges of globalization change how a business operates in different ways. When companies decide to go global, they must be ready and willing to change internal processes. This helps to accommodate new markets and make their global workforce feel comfortable and accepted at work. Companies see many aspects of their businesses change once they enter the global marketplace.

How Globalization Changes Your Daily Businesses Operations

1. Global Communication Challenges

Before starting to branch out from headquarters, firms have to put an established internal communication plan in place since global employees likely work in a different time zone and have a different native language.

Software and other digital tools help smooth global communication hurdles and allows teams to connect easily. Zoom and Google all provide valuable tools for companies trying to manage employees in multiple offices, countries, and time

How Globalization Changes Your Daily Businesses Operations

2. International Employee Expectations

Foreign employees have different expectations when it comes to things like salary and benefits, as well as how they manage their daily work schedules. Companies that want to take advantage of globalization and hire foreign workers need to accommodate them as much as possible. HR teams must also ensure their offers are competitive and on-par with local expectations during the hiring process.

How Globalization Changes Your Daily Businesses Operations

3. Supporting Foreign Customers

Similar to communication changes with employees, companies must also plan for how they run customer service and support in new countries. Customers in the new market where you offer your products or services might not speak your native language or be close to your time zone.

How Globalization Changes Your Daily Businesses Operations

4. Increased Competition

International companies have to adjust more than internal operations. Going global opens up new revenue streams and increases availability to talent. Because of these attractive benefits, and the ease of going global due to services like International PEO, the global marketplace is competitive. As globalization becomes the norm, many companies often seek the same foreign markets which increases competition for businesses.

How Globalization Changes Your Daily Businesses Operations

5. Marketing and Communication Changes

Just like hiring employees in different countries creates internal communication challenges, marketing your products or services to a completely new audience creates obstacles for companies. Businesses need to adjust their marketing strategies to communicate the benefits of their product.

Globalization Challenges and Opportunities:

A Flattened World The Emerging Digital Firm

Why Information Systems Matter

Let's start by examining why information systems and information technology (IT) are so important. There are four reasons why IT will make a difference to you as a manager throughout your career.

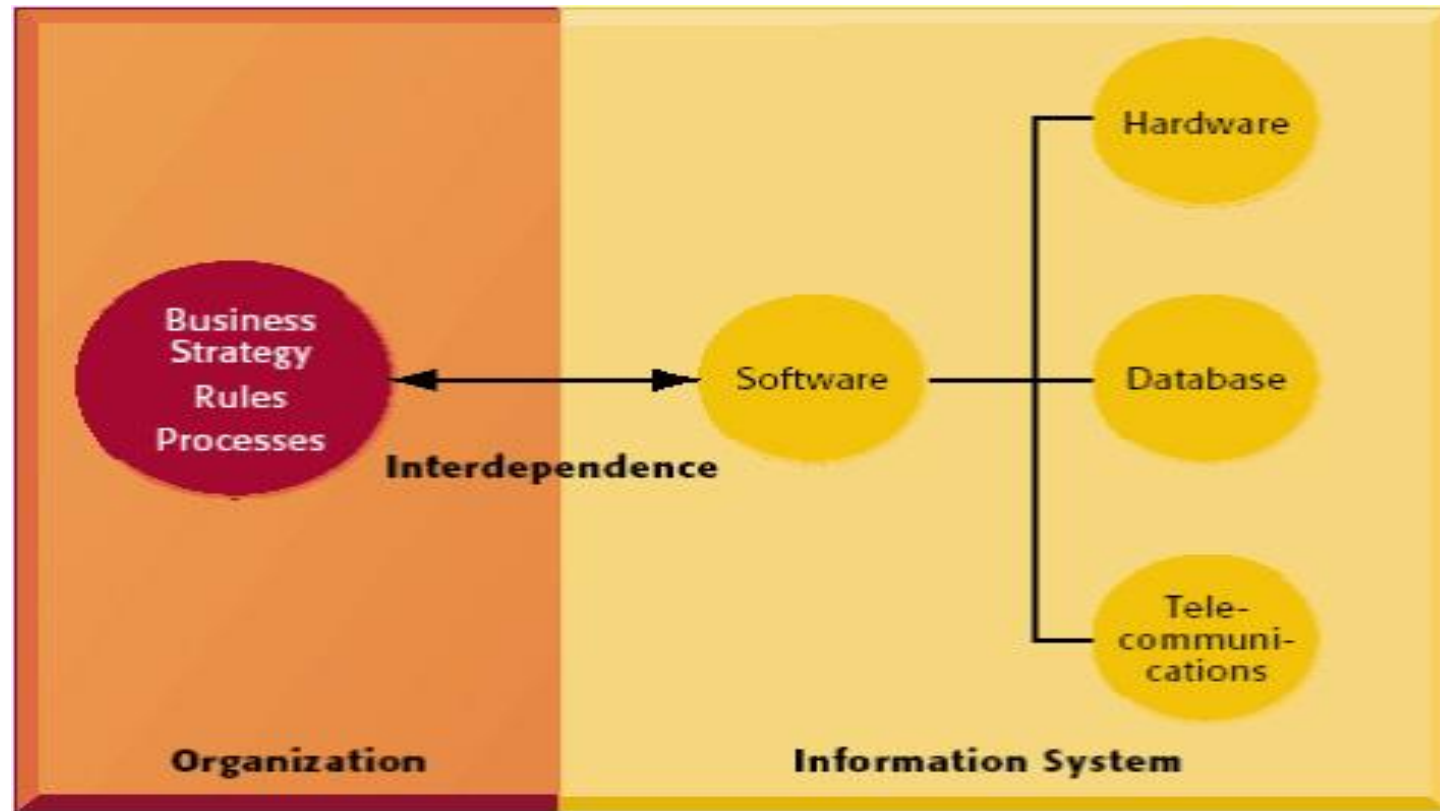
- Capital Management
- Foundation of doing business
- Productivity
- Strategic opportunity and advantage

Globalization Challenges and Opportunities:

Why IT Now? Digital Convergence and the Changing Business Environment

A combination of information technology innovations and a changing domestic and global business environment makes the role of IT in business even more important for managers than just a few years ago. The Internet revolution is not something that happened and then burst but instead has turned out to be an ongoing, powerful source of new technologies with significant business implications for much of this century.

Globalization Challenges and Opportunities:



Globalization Challenges and Opportunities:

Dimensions of Information System

The **dimensions of information systems** include organizations, management, and **information** technology. The key elements of an organization are its people, structure, business processes, politics and culture. An organization coordinates work through a structured hierarchy and formal standard operating procedures.

Dimensions of Information System



Dimensions of Information System

- **Organizational Dimension**

In organizational dimensions, management understands what is embodied in its information system relating organizational stuff like culture, norms and values, core tasks of organization, hierarchy of organization etc. Management should understand that all decisions that we made depend on the information which is provided by information systems. Organizational dimensions of any organization are easy to understand for manager because it is his line and he is better aware of the culture and norms of his organization.

Dimensions of Information System

- **Management Dimension**

In management dimension of information systems, information systems assist manager to allocate resources, hire staff, motivate staff, realize risky areas and allocate experienced staff of such risky areas. Information system assists manager to see different designs of products and to make new products. Manager use different techniques from information system to reduce the cost of its product like; target costing technique, Organization produces its reports, financial statements and forecasts by the help of information systems.

Dimensions of Information System

- Technological dimension

The **technological dimension** of culture is its capital, its tools and skills, and ways of dealing with the physical environment. It is the interface between humanity and nature. Technology is much a cultural **dimension** as beliefs and patterns of interaction. It is symbolic.

Impact of IT in Business Firm

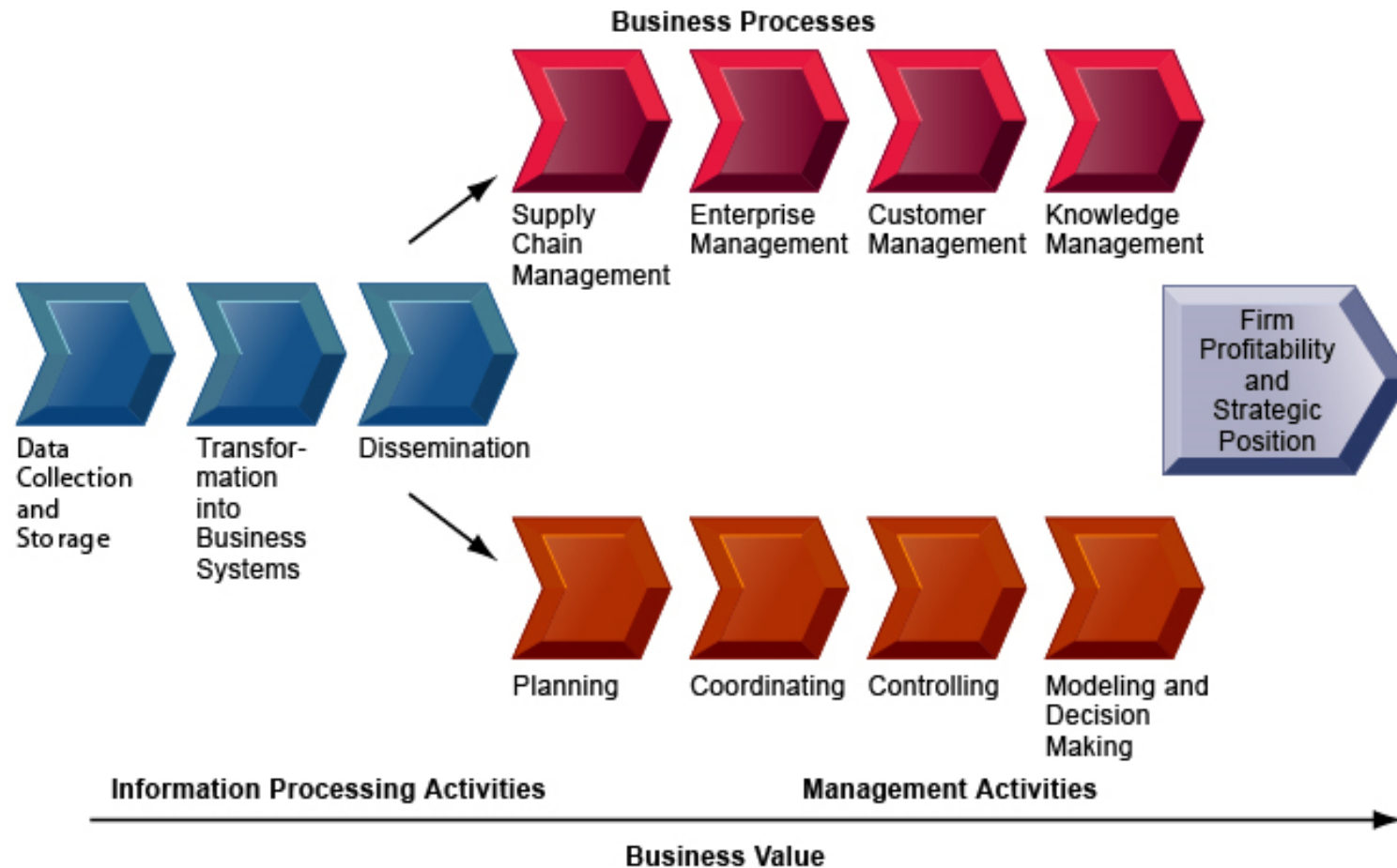
There are five factors to consider when assessing the growing impact of IT in business firms both today and over the next ten years.

- Internet growth and technology convergence
- Transformation of the business enterprise
- Growth of a globally connected economy
- Growth of knowledge and information-based economies
- Emergence of the digital firm

Impact of IT in Business Firm

From a business perspective, an information system is an important instrument for creating value for the firm. Information systems enable the firm to increase its revenue or decrease its costs by providing information that helps managers make better decisions or that improves the execution of business processes. Every business has an information value chain in which raw data is systematically acquired and then transformed through various stages that add value to that information.

Business perspective of information system



Business perspective of information system

The business perspective calls attention to the organizational and managerial nature of information systems. An information system represents an organizational and management solution based on information technology to a Challenge or problem posed by the environment. Some firms achieve better results from their information systems than others. Studies of returns from information technology investments show that there is considerable variation in the returns firms receive.

Complementary assets

Complementary investments include:

- Organizational assets:

These include a supportive business culture that values efficiency and effectiveness, an appropriate business model, efficient business processes, decentralization of authority, highly distributed decision rights and a strong information system (IS) development team.

Complementary assets

Assets required to derive value from a primary investment. For instance, to realize value from automobiles requires complementary investments in highways, roads, gasoline stations, repair facilities and a legal regulatory structure to set standards and control drivers.

Complementary assets

- **Managerial assets:**

These include strong senior management support for change, incentive systems that monitor and reward individual innovation, an emphasis on teamwork and collaboration, training programs and a management culture that values flexibility and knowledge.

Complementary assets

- **Social assets:**

These are not made by the firm but by the society at large, other firms, governments, and other key market actors, such as the Internet, educational systems, network and computing standards, regulations and laws and the Presence of technology and service firm.

Complementary assets

Research indicates that firms that support their technology investments with investments in complementary assets, such as new business processes or training, receive superior returns.

Hands on MIS projects

1. Forecasting, planning, and controlling of the business process.
2. Numbers of good received can be controlled and monitored.
3. Better customer service.
4. Decision making

Impact of Business

- Business Impact:
- Increase in the rate at which responses are given to customers, this is time consuming problem.
- The flow of information may drop the number of customers as the required information takes long before it is gotten.
- Work activity not organized well could bring inefficiency and reduce in overall performance of the company. Business process is not well designed and coordinated because there is no competitive strength.

Improving impact of business by using IT

Describe how this process could be improved with information technology.

- Improvement in performance would increase competitive advantages.
- Some business solutions such as reducing cost and improving the service are necessary.
- Organization of work activities using information technology will increase the overall performance of the business.
- Better performance and recording of daily transactions are necessary to conduct business in website.

Improving impact of business by using IT

- **Management Decision Problem**

A **Management Decision Problem** is where by when a situation arises and **management** of a company needs to make a **decision** which requires research and Hence starts the research process.

Management problem are actually the action oriented one.

Research problem: It is about the data collection ,sampling and analysis of data and then draw the result. The **research** problems mainly focuses on the causes only.

Improving impact of business by using IT

what is a management problem?

Problem management is the process responsible for **managing** the life cycle of all **problems** that happen or could happen in an IT service. The Information Technology Infrastructure Library defines a **problem** as the cause of one or more incidents.

Improving impact of business by using IT

What is decision problem in marketing research?

Decision Problem in **marketing research**, the basic question to which a client organization seeks an answer from a **marketing research** agency. in conducting the **research**, the agency may break the **decision problem** down into any number of smaller **research** questions in order to provide an answer.

Improving decision making

Get real-time data for better decision making. Try ProjectManager.com and get online PM tools that can help you spot problems, build solutions and steer successful projects.

Managers spend a lot of time making decisions, big and small. And we cannot be right all the time. But what we do want is to be able to make better decisions every time. But how do we improve decision-making? We first need to know what “good” looks like.

Improving decision making

Consider a decision to be good if:

- The right person took it
- With proper assessment of the best available evidence.
- Following a sound process.

Improving decision making

9 TIPS FOR BETTER DECISION-MAKING



1

Learn from Experience

2

Remain Skeptical

3

Have Options

4

Argue It Out

5

Know the Context

6

Experiment

7

Trash It

8

Assume It's Wrong

9

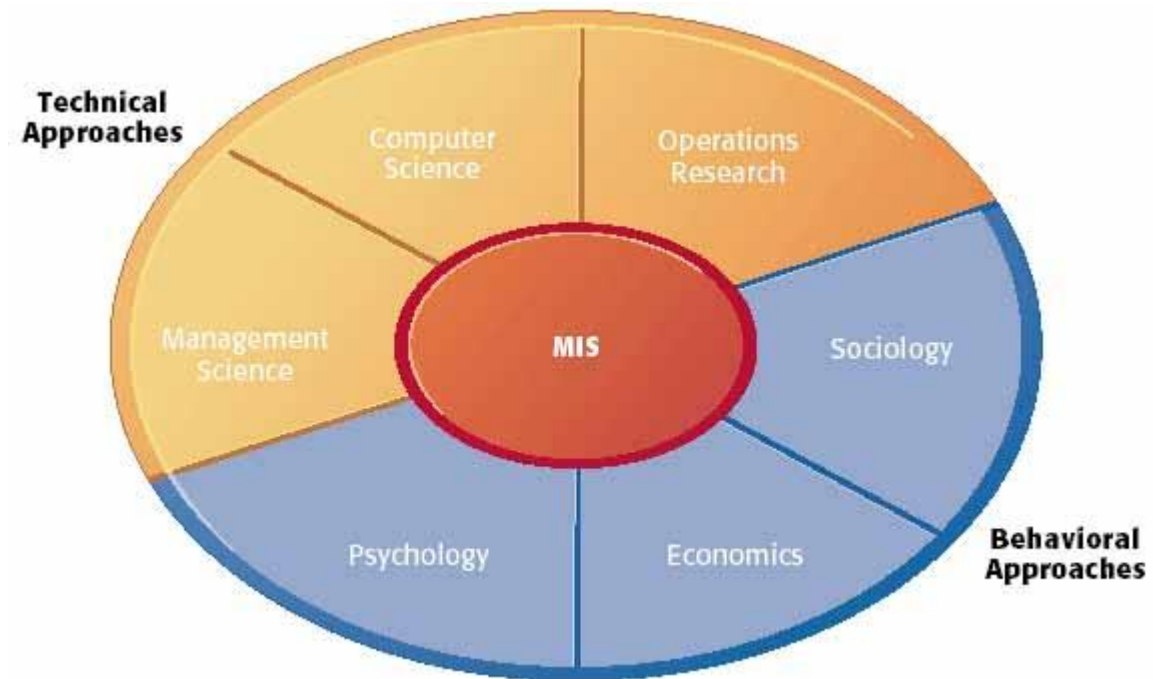
Get Others' Perspectives



Academic Disciplines used to Study Information Systems

- List and describe each discipline that contributes to a technical approach to information systems.
- List and describe each discipline that contributes to a behavioral approach to information systems.
- Describe the sociotechnical perspective on information systems.

Academic Disciplines used to Study Information Systems



Academic Disciplines used to Study Information Systems

Technical Approach

The technical approach to information systems emphasizes mathematically based models to study information systems as well as the physical technology and formal capabilities of these systems. The disciplines that contribute to the technical approach are computer science, management science and operations research.

Academic Disciplines used to Study Information Systems

Behavioral Approach

An important part of the information systems field is concerned with behavioral issues that arise in the development and long-term maintenance of information systems. Issues such as strategic business integration, design, implementation, utilization and management cannot be explored usefully with the models used in the technical approach. Other behavioral disciplines contribute important concept and methods.

Sociotechnical Systems

A socio-technical system (STS) is one that considers requirements spanning hardware, software, personal and community aspects. It applies an understanding of the social structures, roles and rights (the social sciences) to inform the design of systems that involve communities of people and technology.

Sociotechnical Systems

